

Q.22 Write various effects of force.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw stress strain curve for a ductile material subjected to tensile load and explain its main parts.

Q.24 Explain

- a) Free body diagram
- b) Triangle law of forces

Q.25 A load of 200N is raised by means of a simple machine through a distance of 100 cm. If effort applied is 40N and it moves through a distance of 600 cm, find mechanical advantage, velocity ratio and efficiency of machine.

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2nd Sem / Mechanical (Tool & die Design)

Subject : Mechanics of Solids

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Lami's theorem can be applied when the body is-

- a) Vibrating b) Moving
- c) Rotating d) In equilibrium

Q.2 Moment of a force-

- a) Varies directly with its distance from the pivot.
- b) Varies inversely with its distance from the pivot.
- c) Independent of its distance from the pivot.
- d) None of the above

Q.3 The coefficient of friction depends on-

- a) Area of contact b) Shape of surface
- c) Nature of surface d) Strength of surface

Q.4 The young's Modulus of elasticity is the ratio of

- a) Longitudinal stress and longitudinal strain
- b) Lateral stress and lateral strain.
- c) Volumetric stress and volumetric strain.
- d) None of the above

Q.5 The unit of moment of inertia of an area is-

- a) Kg-m^2
- b) m^4
- c) m^3
- d) kg-m/s^2

Q.6 A beam is said to be of uniform strength, if

- a) Bending Moment is same throughout the beam
- b) Shear stress is the same through the beam
- c) Deflection is the same throughout the beam
- d) Bending stress is the same at every section along its length.

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The force of attraction between any two bodies is called_____

Q.8 Define Strain

Q.9 An ideal machine is one in which output=

Q.10 The thermal stress in a bar free to extend when temperature is increases, is_____

Q.11 Angle of friction is always_____ than 90° .

Q.12 Define mechanical advantage.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Drive the relation between mechanical advantage, velocity ratio and efficiency of a machine.

Q.14 Define resilience and proof resilience.

Q.15 Find the resultant of the forces of 40N and 60N when the are at an angle of 60° with each other.

Q.16 Define moment. Differentiate between clockwise and anticlockwise moments.

Q.17 How does a lubricant reduce friction?

Q.18 Find the centroid of a L plane section having length 100 mm and breadth 80mm and thickness of 20 mm.

Q.19 Explain different types of strain.

Q.20 Explain various assumptions made during deriving torsion equation.

Q.21 Explain working principle and applications of wheel and axle.